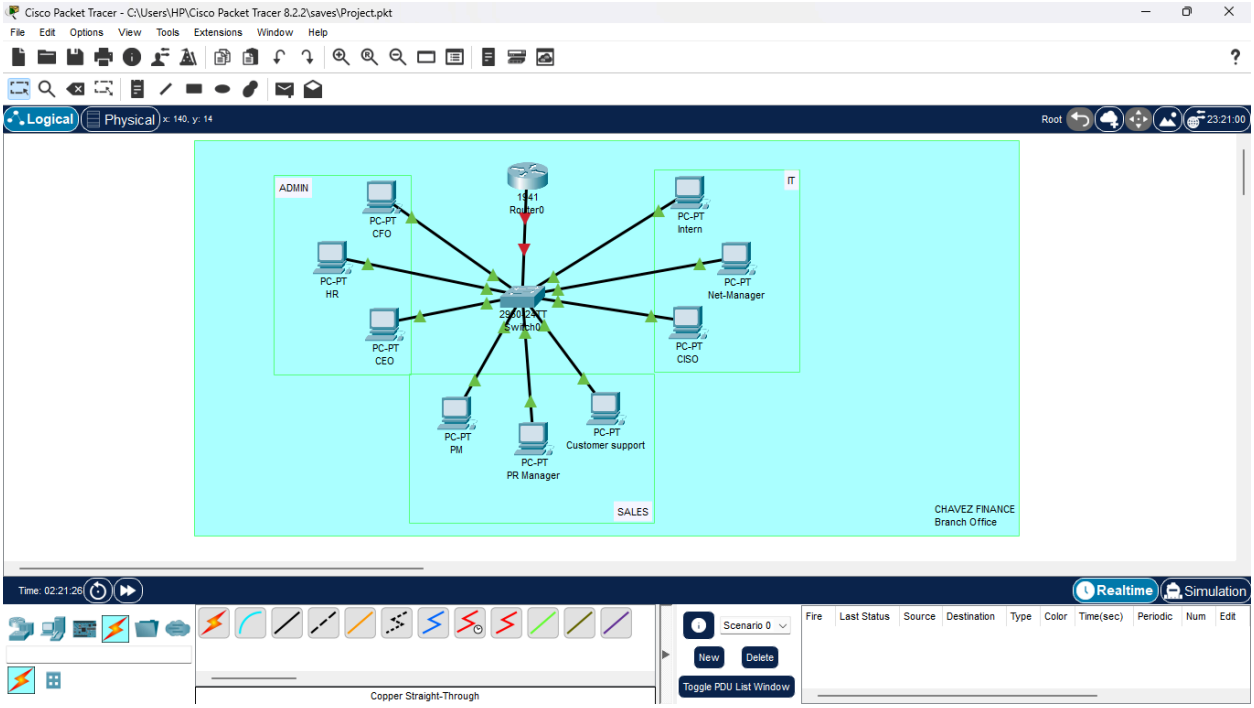
NETWORK SIMULATION

Scenario; I have recently been hired as a Network Security Engineer Intern at CHAVEZ FINANCE.

The company has requested that I design and implement a secure small enterprise network using Cisco Packet Tracer to simulate its branch office infrastructure.

Network Topology Diagram showing devices, connections, and VLAN assignments.

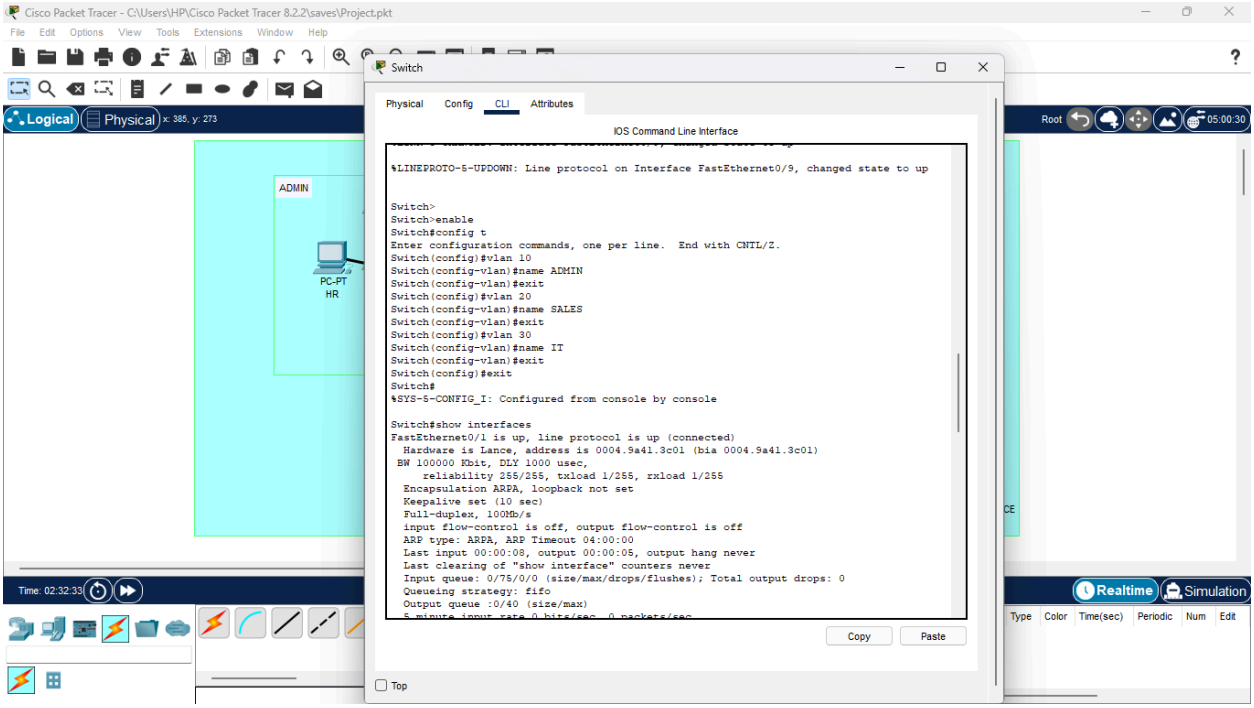
This is the representation of the Network STAR topology of CHAVEZ FINANCE Enugu Branch Office.
The network is segmented into three VLANs; ADMIN, SALES and IT, with three PCs each. The number of PCs is very flexible and can be increased or decreased.
The network has a build of ;
1. A router which facilitates inter-vlan routing. Router 1941
2. A central switch that facilitates each PC's connection to the network and segmentation. Switch 2960-24TT
3. PC's which represent employees in their departments
4. Straight-through cables



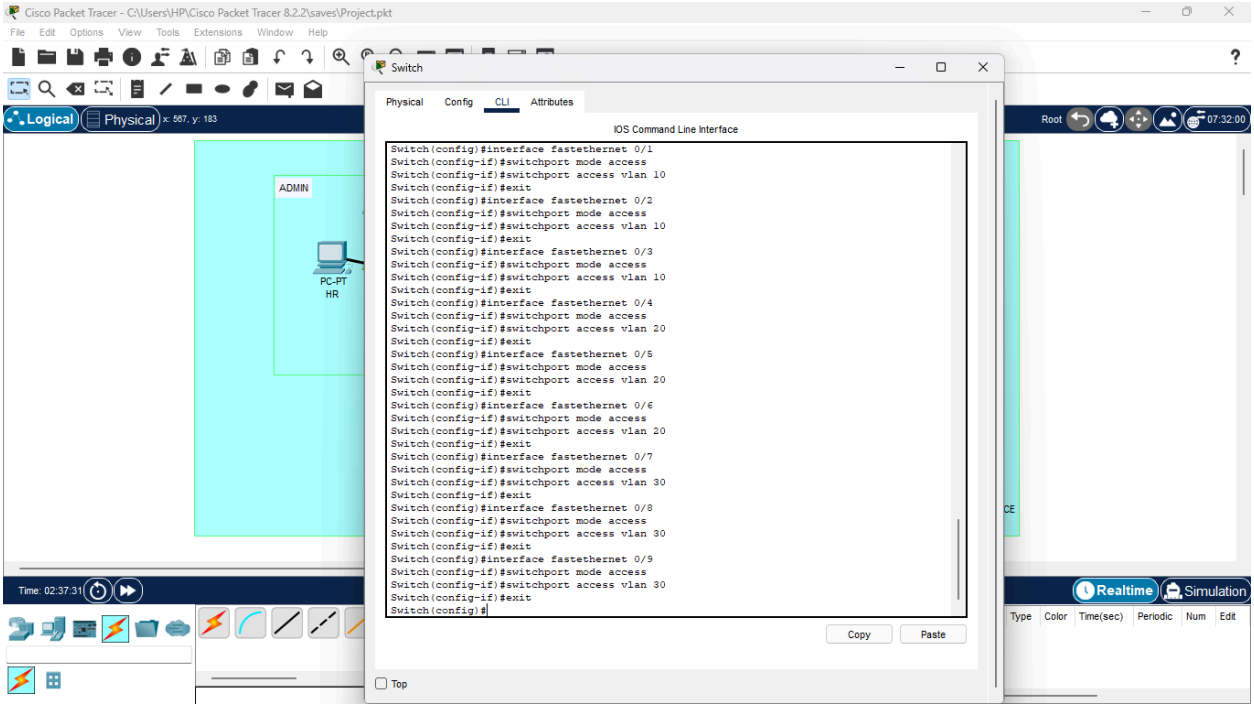
Branch office Network Topology Diagram

1. Network Design & VLAN Segmentation

vlan 10
name ADMIN
exit

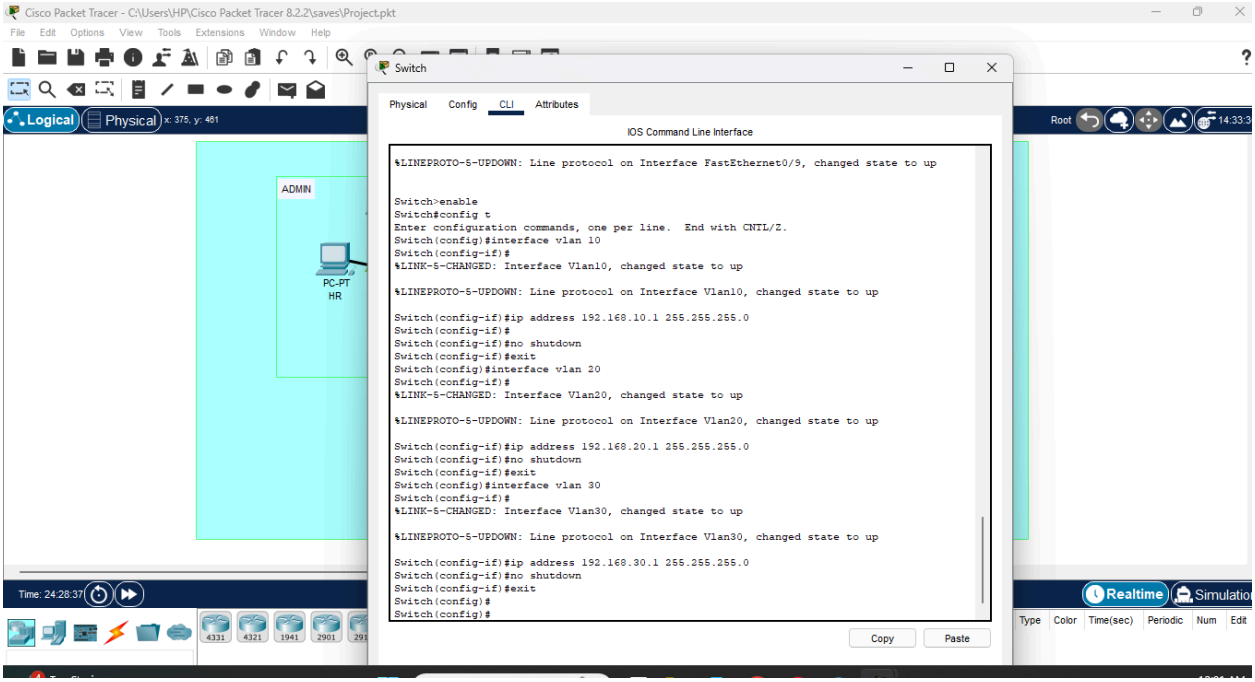


Interface fastEthernet 0/1
switchport mode access
switchport access vlan 10



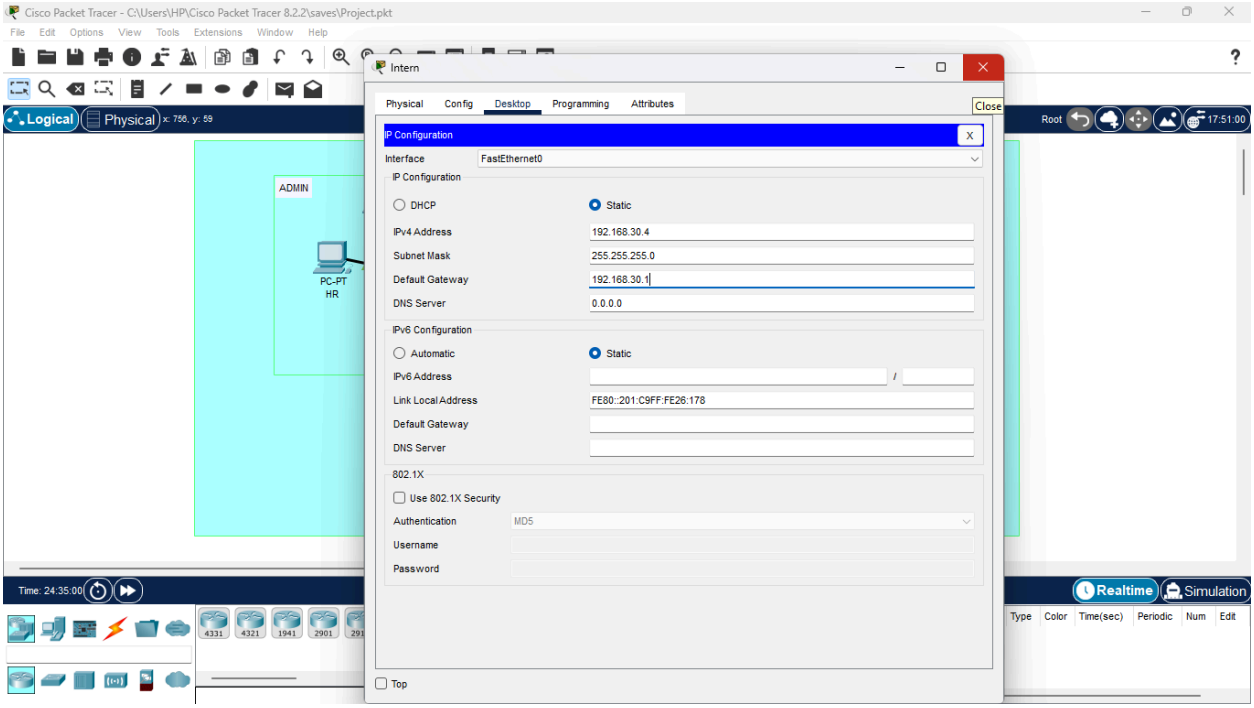
Right here, I assigned FastEthernet ports to specific VLANS
FastEthernet 0/1 — 0/3 for ADMIN VLAN
FastEthernet 0/4 — 0/6 for SALES VLAN
FastEthernet 0/7 — 0/9 for IT VLAN

```
interface vlan 10
ip address 192.168.10.1 255.255.255.0
no shutdown
exit
```

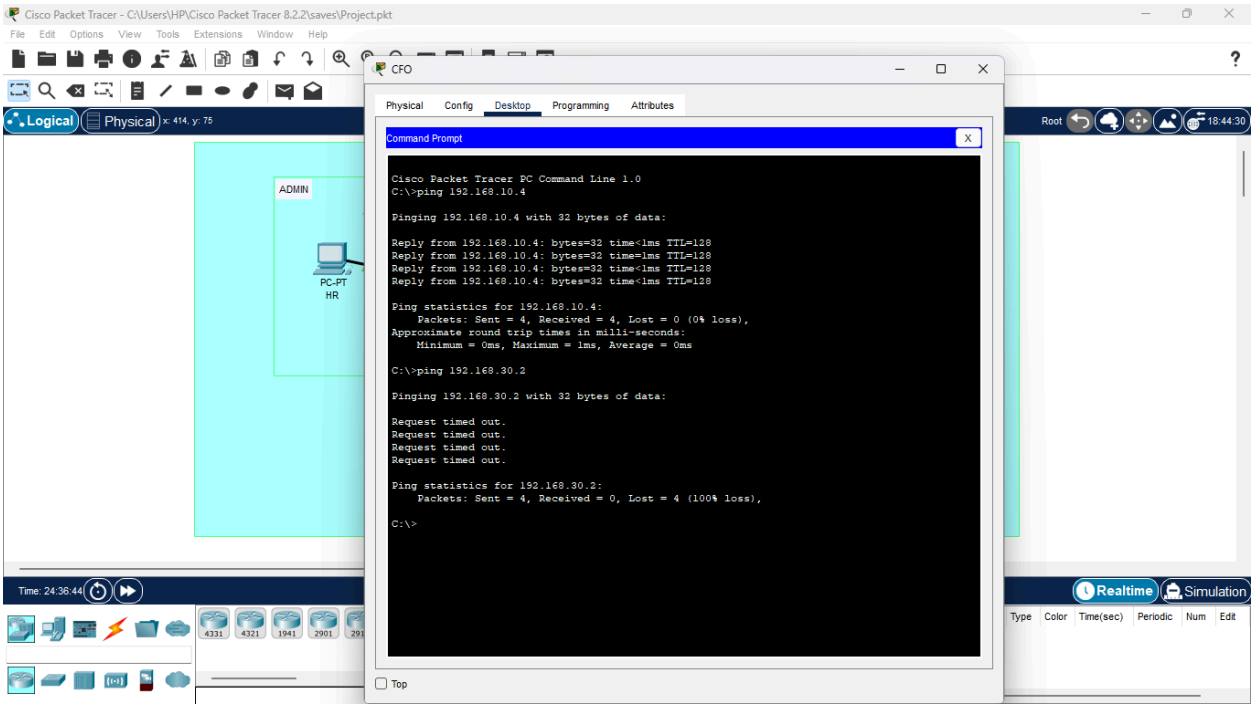


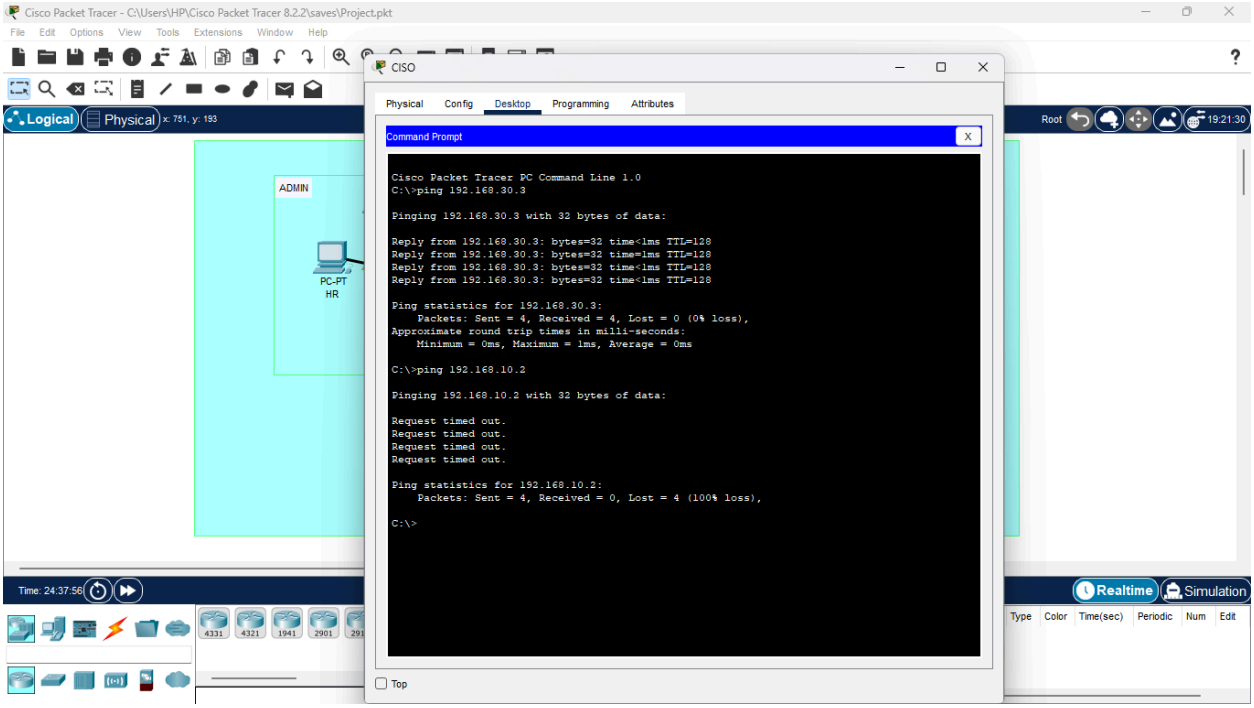
I assigned Subnets to each VLAN
Admin VLAN: VLAN 10 - 192.168.10.0/24
Sales VLAN: VLAN 20 - 192.168.20.0/24
IT VLAN: VLAN 30 - 192.168.30.0/24

I configured each PC on the network with a static IP under each of their specific subnet



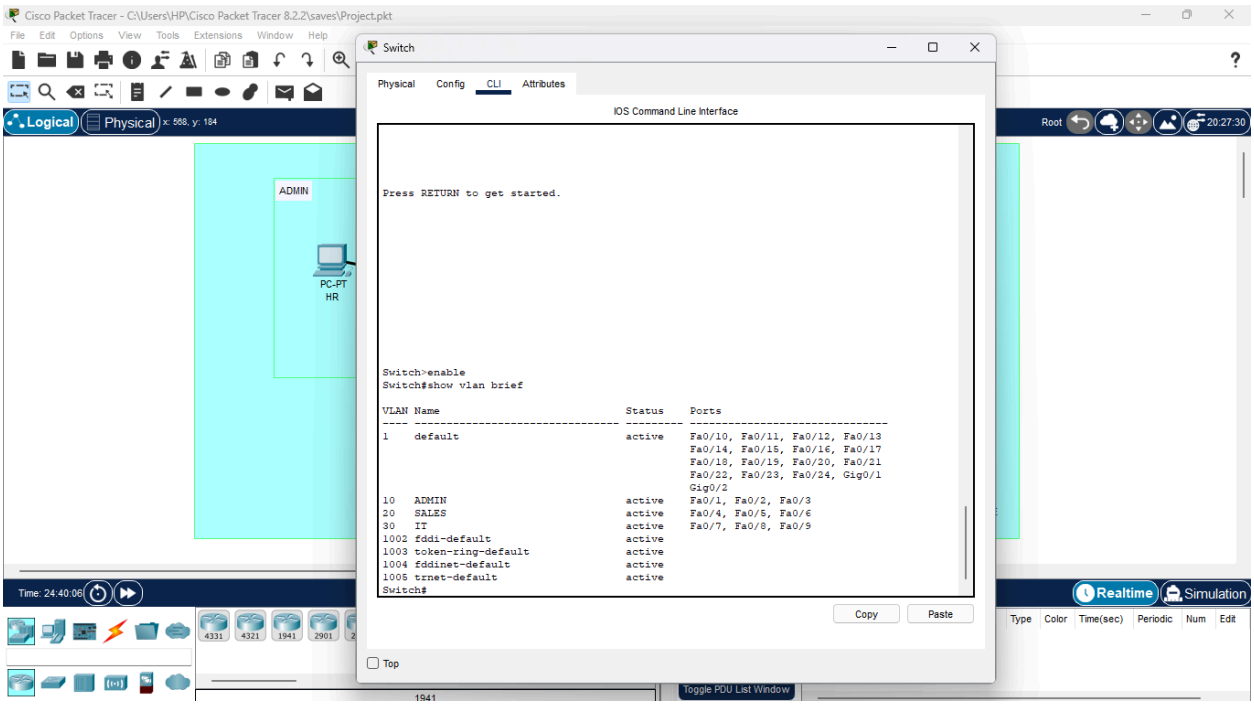
Here, I am troubleshooting the network settings to make sure hosts can interact within their Subnets only by default. Using the “ping” command.





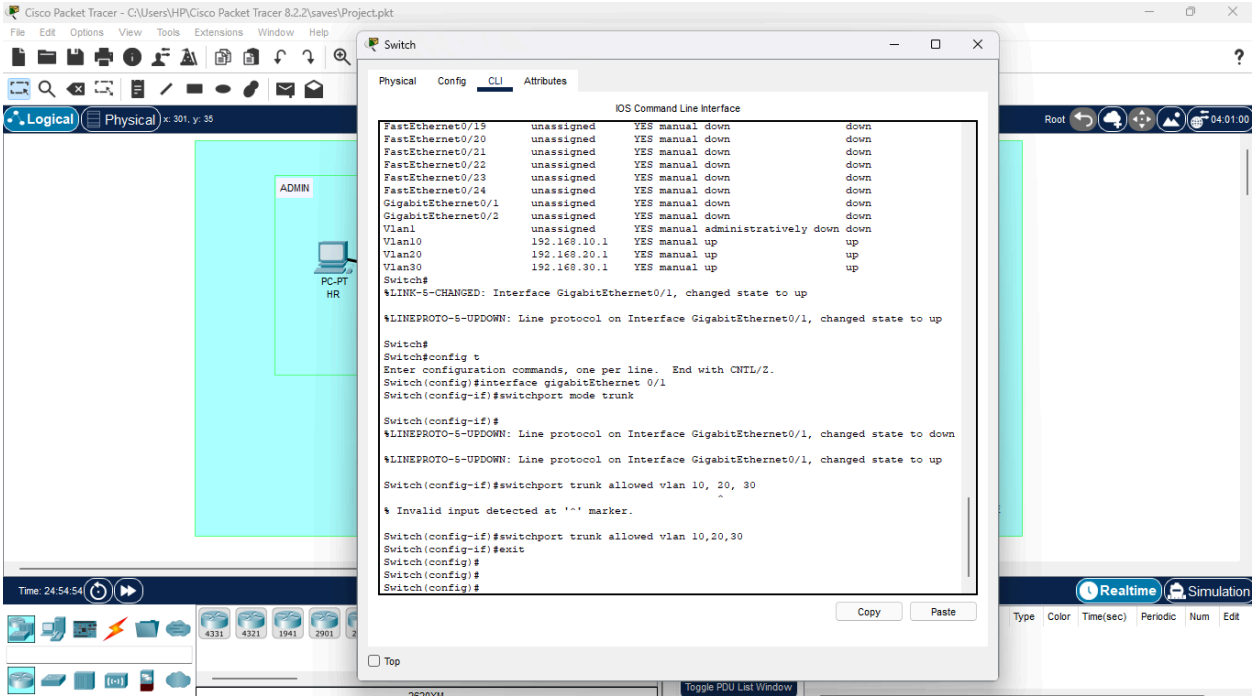
`Here is a confirmation of VLAN segmentation on the switch

show vlan brief



2. Inter-VLAN Routing

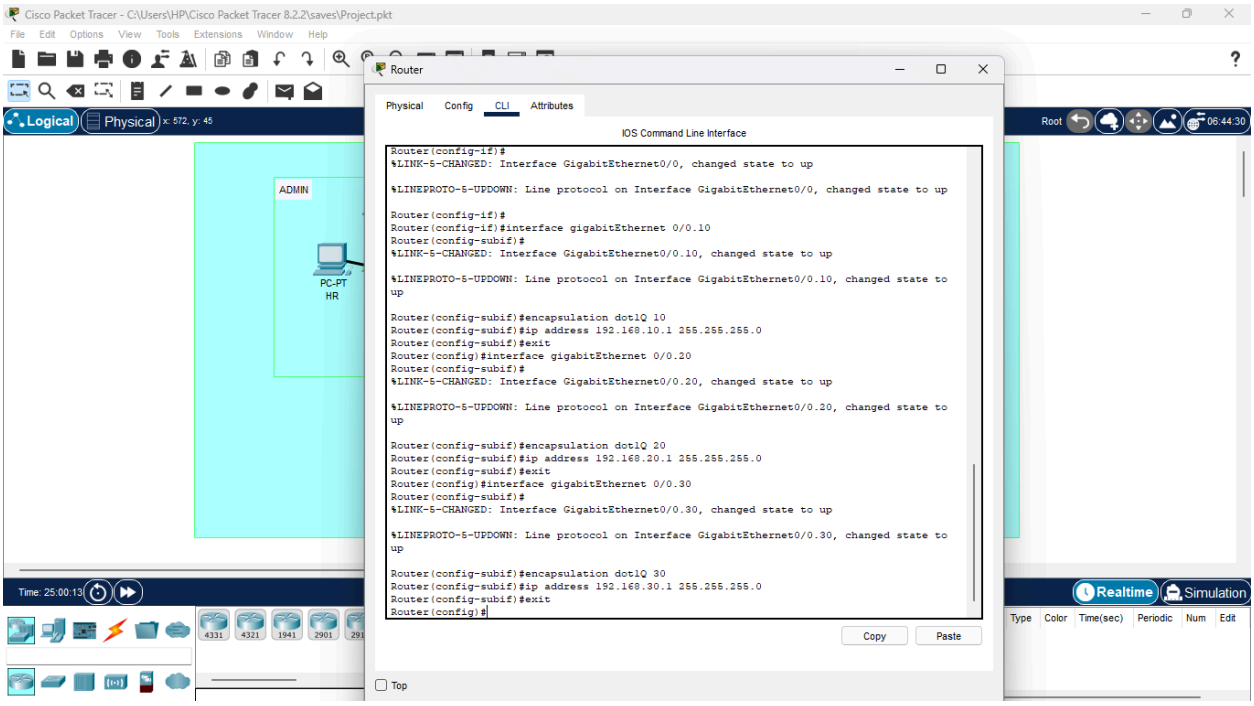
Setting up trunk ports on the switch gigabit interface
interface gigabitEthernet 0/1
switchport mode trunk
switchport allowed vlan 10,20,30
exit



Introduce the Router to assign subnets to each vlan
Configuring the Router gigabit sub-interfaces on the gigabitEthernet 0/0 interface.
Configure "encapsulation dot1Q" on each sub-interface for secured inter VLAN communication
Admin VLAN: VLAN 10 - `192.168.10.0/24`
Sales VLAN: VLAN 20 - `192.168.20.0/24`
IT VLAN: VLAN 30 - `192.168.30.0/24`

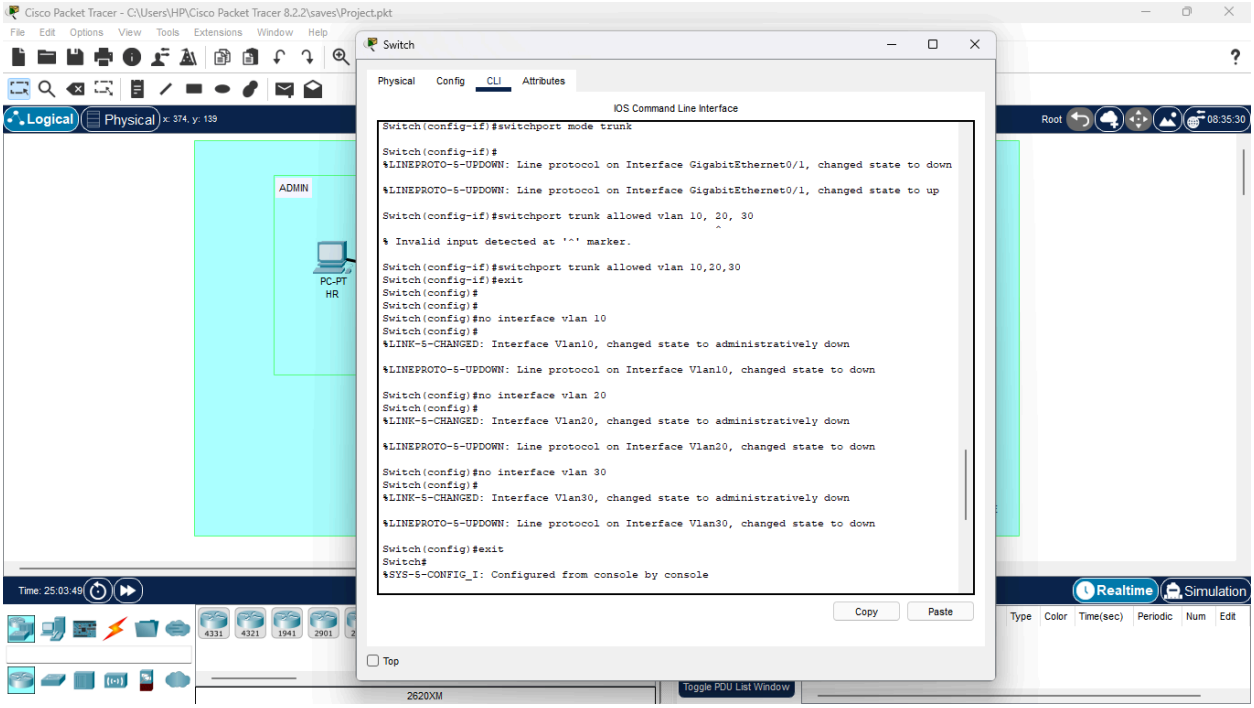
interface gigabitEthernet 0/0.10
encapsulation dot1Q 10

ip address 192.168.10.1 255.255.255.0
exit

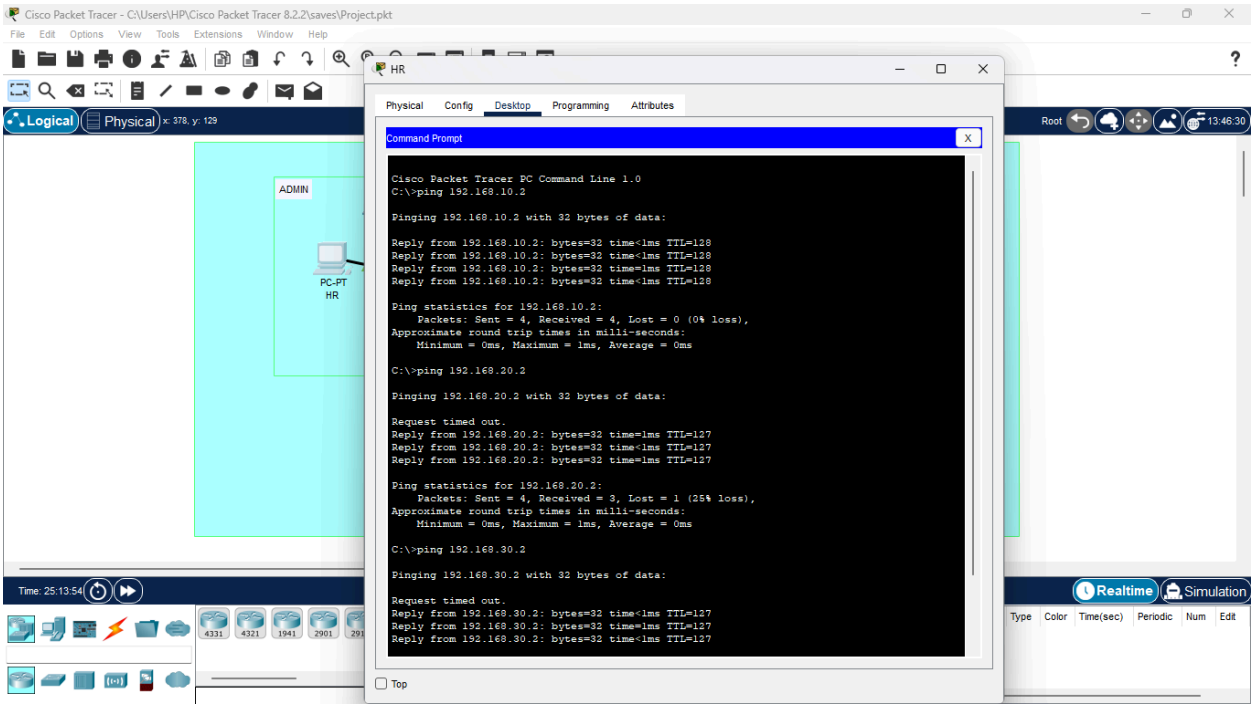


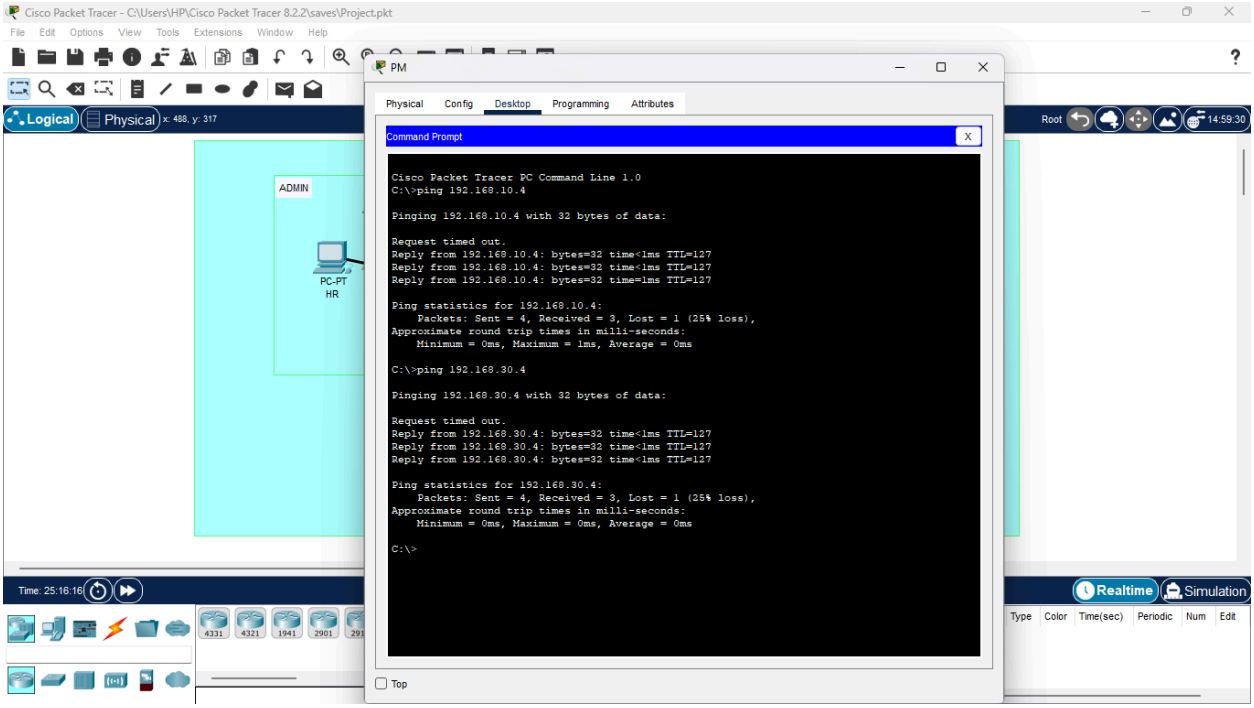
Remove the vlan interfaces on switch to avoid interference with the configuration on the router.

no interface vlan 10



Ensure the routing is functional by pinging PCs within and outside each segment.



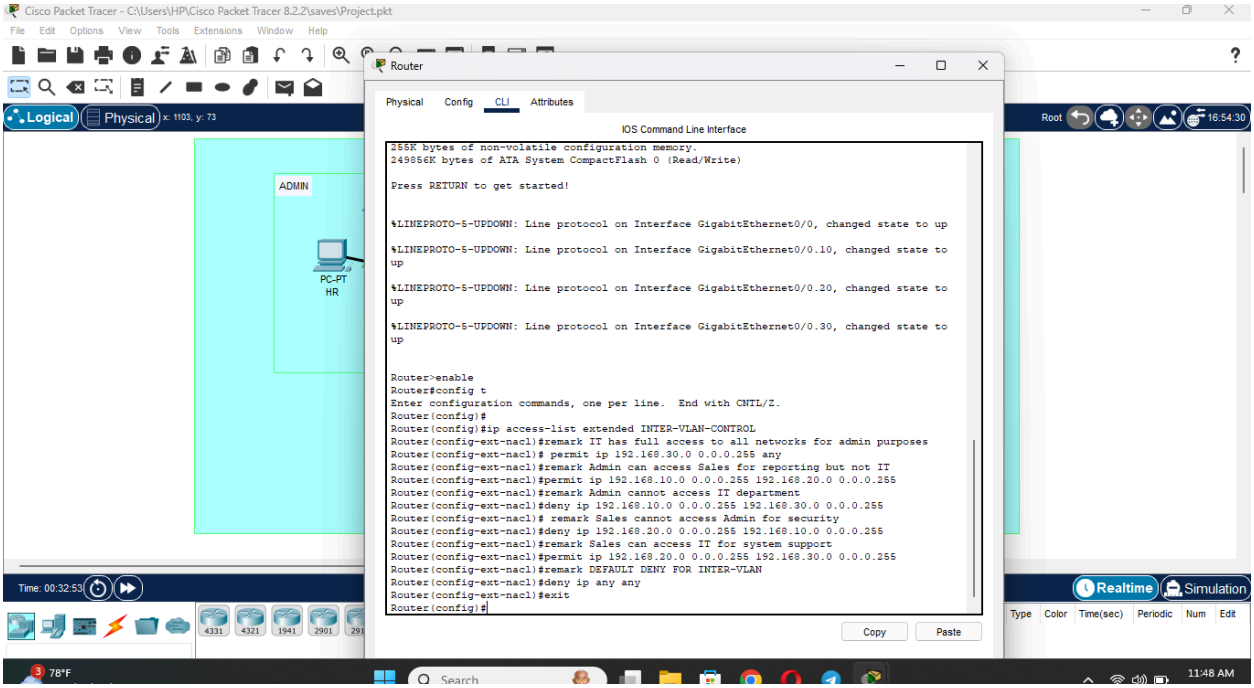


3. **Access Control Lists (ACLs)**

Configuring Access control list on the Router
! Create extended ACL for inter-VLAN traffic control

Router> enable
Router# configure terminal
Router(config)# ip access-list extended INTER-VLAN-CONTROL
Router(config-ext-nacl)# remark IT has full access to all networks for admin purposes
Router(config-ext-nacl)# permit ip 192.168.30.0 0.0.0.255 any
Router(config-ext-nacl)# remark Admin can access Sales for reporting but not IT
Router(config-ext-nacl)# permit ip 192.168.10.0 0.0.0.255 192.168.20.0 0.0.0.255
Router(config-ext-nacl)# remark Admin cannot access IT department
Router(config-ext-nacl)# deny ip 192.168.10.0 0.0.0.255 192.168.30.0 0.0.0.255
Router(config-ext-nacl)# remark Sales cannot access Admin for security
Router(config-ext-nacl)# deny ip 192.168.20.0 0.0.0.255 192.168.10.0 0.0.0.255
Router(config-ext-nacl)# remark Sales can access IT for system support

Router(config-ext-nacl)# permit ip 192.168.20.0 0.0.0.255 192.168.30.0 0.0.0.255
Router(config-ext-nacl)# remark DEFAULT DENY FOR INTER-VLAN
Router(config-ext-nacl)# deny ip any any
Router(config-ext-nacl)# exit



Create Additional ACLs for Specific Protocols (Enhanced Security)

Create ACL for IT management protocols only

Router(config)# ip access-list extended IT-MANAGEMENT-ACL
Router(config-ext-nacl)# remark Allow IT to use all protocols for management
Router(config-ext-nacl)# permit ip 192.168.30.0 0.0.0.255 any
Router(config-ext-nacl)# exit

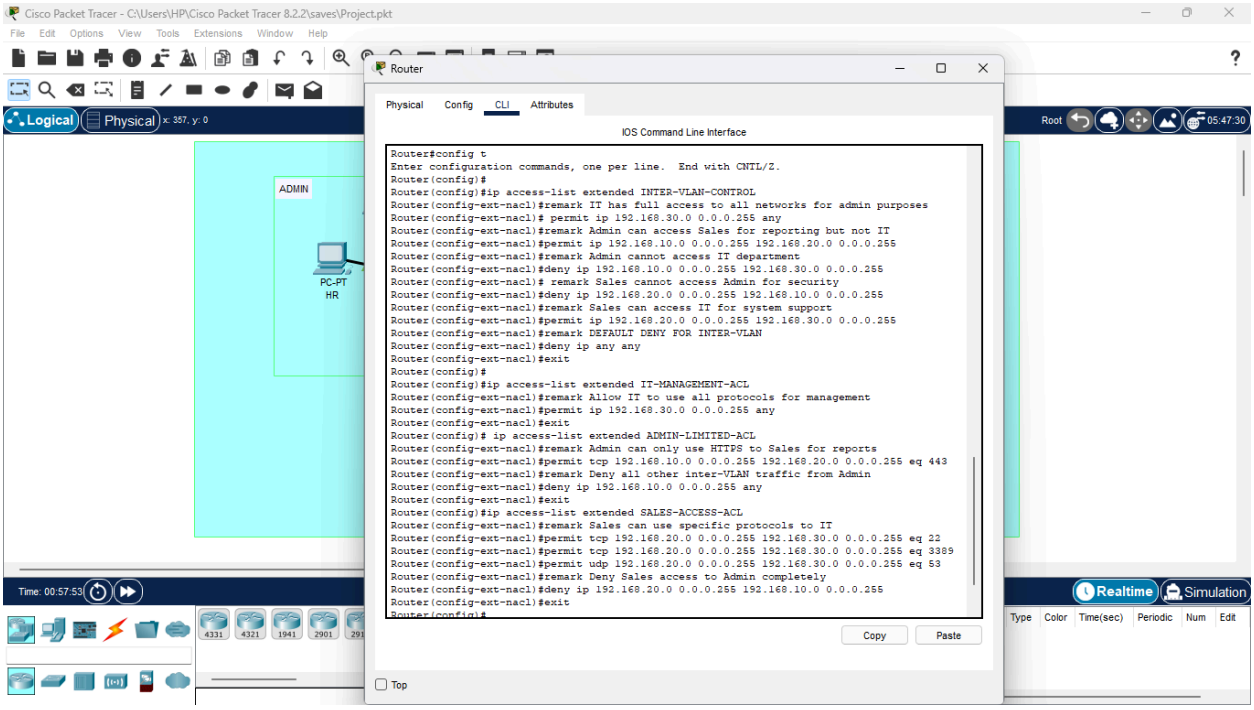
Create ACL for Admin limited access

Router(config)# ip access-list extended ADMIN-LIMITED-ACL
Router(config-ext-nacl)# remark Admin can only use HTTP/HTTPS to Sales for reports
Router(config-ext-nacl)# permit tcp 192.168.10.0 0.0.0.255 192.168.20.0 0.0.0.255 eq 443
Router(config-ext-nacl)# remark Deny all other inter-VLAN traffic from Admin

Router(config-ext-nacl)# deny ip 192.168.10.0 0.0.0.255 any
Router(config-ext-nacl)# exit

Create ACL for Sales department

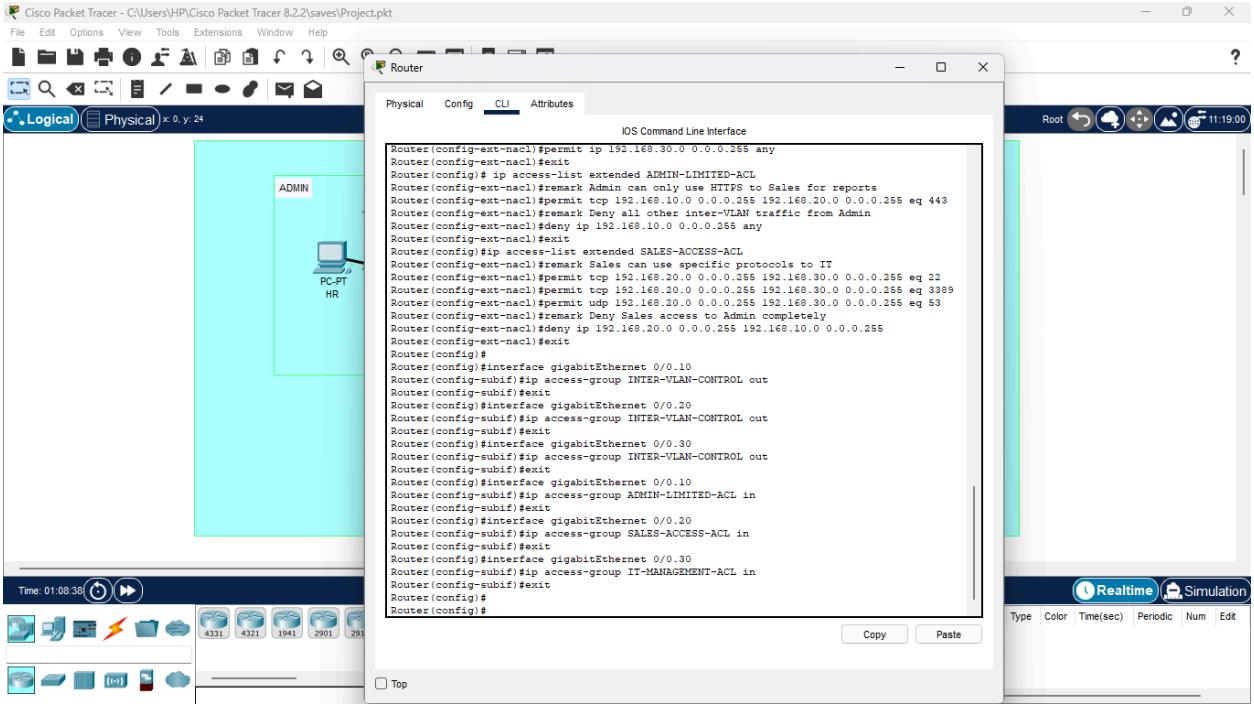
Router(config)# ip access-list extended SALES-ACCESS-ACL
Router(config-ext-nacl)# remark Sales can use specific protocols to IT
Router(config-ext-nacl)# permit tcp 192.168.20.0 0.0.0.255 192.168.30.0 0.0.0.255 eq 22
Router(config-ext-nacl)# permit tcp 192.168.20.0 0.0.0.255 192.168.30.0 0.0.0.255 eq 3389
Router(config-ext-nacl)# permit udp 192.168.20.0 0.0.0.255 192.168.30.0 0.0.0.255 eq 53
Router(config-ext-nacl)# remark Deny Sales access to Admin completely
Router(config-ext-nacl)# deny ip 192.168.20.0 0.0.0.255 192.168.10.0 0.0.0.255
Router(config-ext-nacl)# exit



Assign ACLs to corresponding VLAN out-bound and in-bound traffic

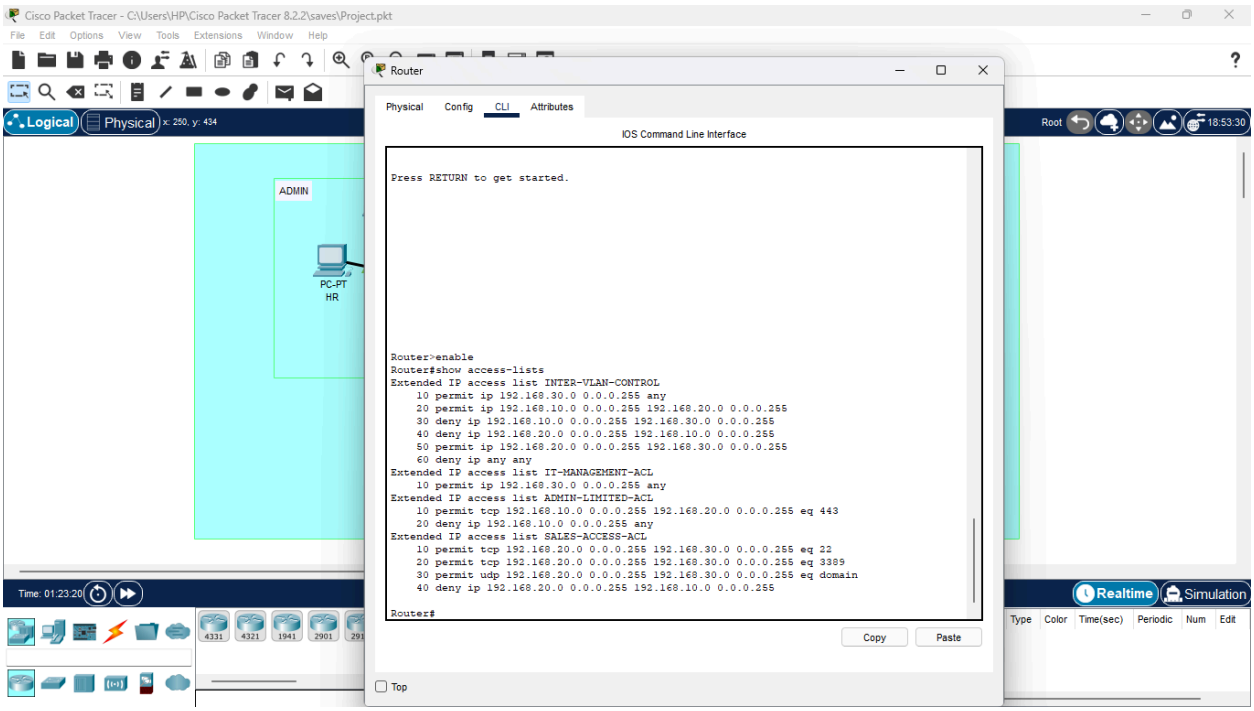
Apply ACLs to Router Subinterfaces

```
! Apply to Admin VLAN subinterface - OUTBOUND direction
Router(config)# interface gigabitEthernet 0/0.10
Router(config-subif)# ip access-group INTER-VLAN-CONTROL out
Router(config-subif)# exit

! Apply to Sales VLAN subinterface - OUTBOUND direction

Router(config)# interface gigabitEthernet 0/0.20
Router(config-subif)# ip access-group INTER-VLAN-CONTROL out
Router(config-subif)# exit

! Apply to IT VLAN subinterface - OUTBOUND direction
Router(config)# interface gigabitEthernet 0/0.30
Router(config-subif)# ip access-group INTER-VLAN-CONTROL out
Router(config-subif)# exit

! Apply to Admin VLAN subinterface - INBOUND direction
Router(config)# interface gigabitEthernet 0/0.10
Router(config-subif)# ip access-group ADMIN-LIMITED-ACL in
Router(config-subif)# exit

! Apply to Sales VLAN subinterface - INBOUND direction

Router(config)# interface gigabitEthernet 0/0.20
Router(config-subif)# ip access-group SALES-ACCESS-ACL in
Router(config-subif)# exit

! Apply to IT VLAN subinterface - INBOUND direction
Router(config)# interface gigabitEthernet 0/0.30
Router(config-subif)# ip access-group IT-MANAGEMENT-ACL out
Router(config-subif)# exit
```



Ensure the ACL list is configured and saved

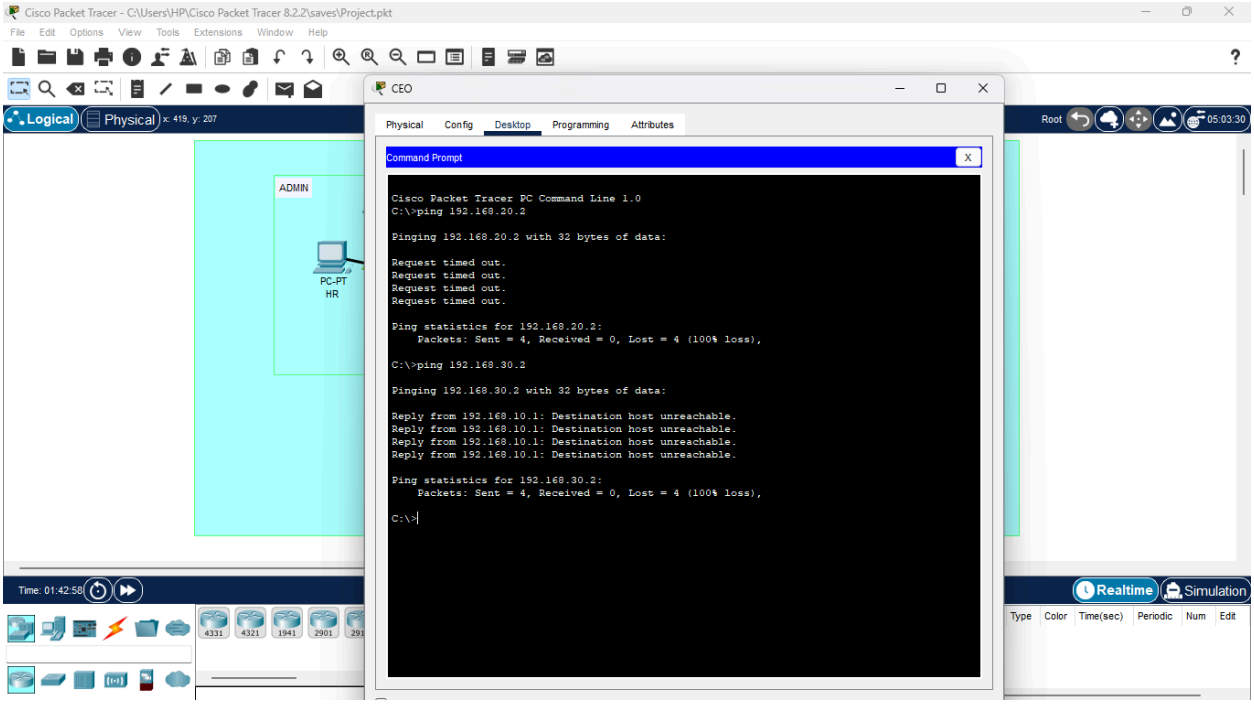
show access-lists



Troubleshooting ACLs functionality:

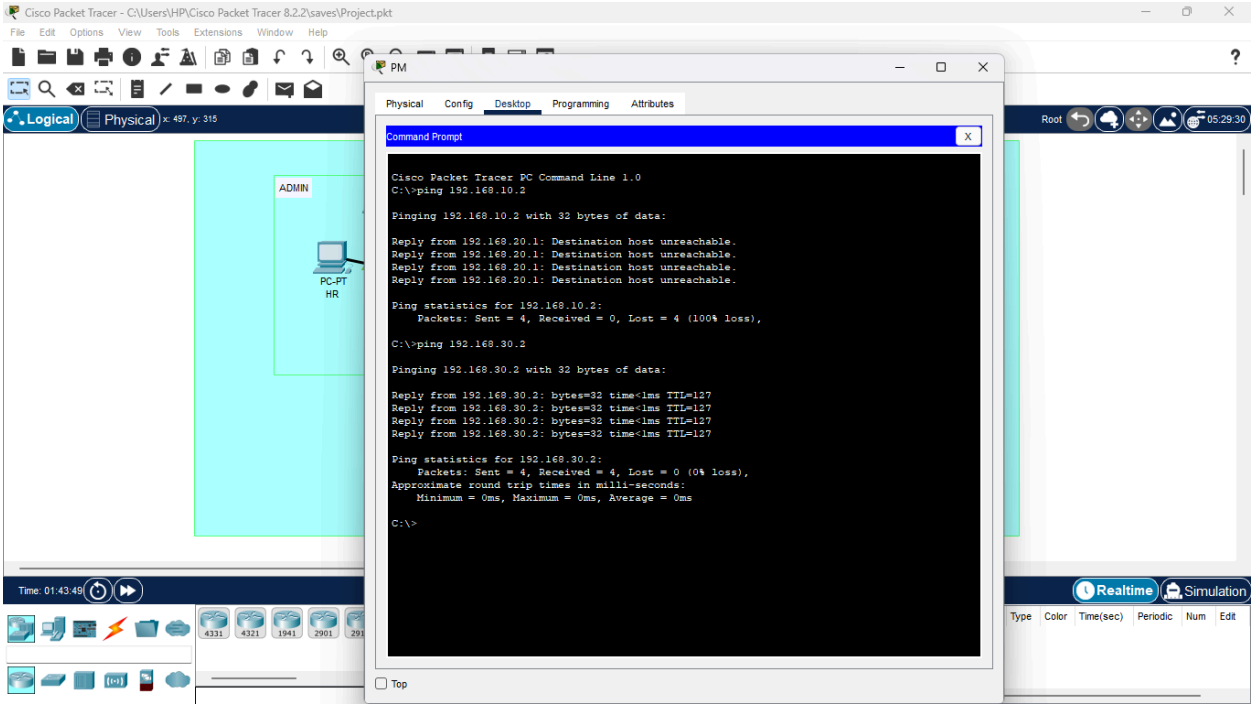
From Admin to sales Request timed out because Admin rules allows traffic to Sales but Sales' outbound rules denies response to Admin.

From Admin to IT Destination unreachable because Admin's outbound rules denies traffic to IT.



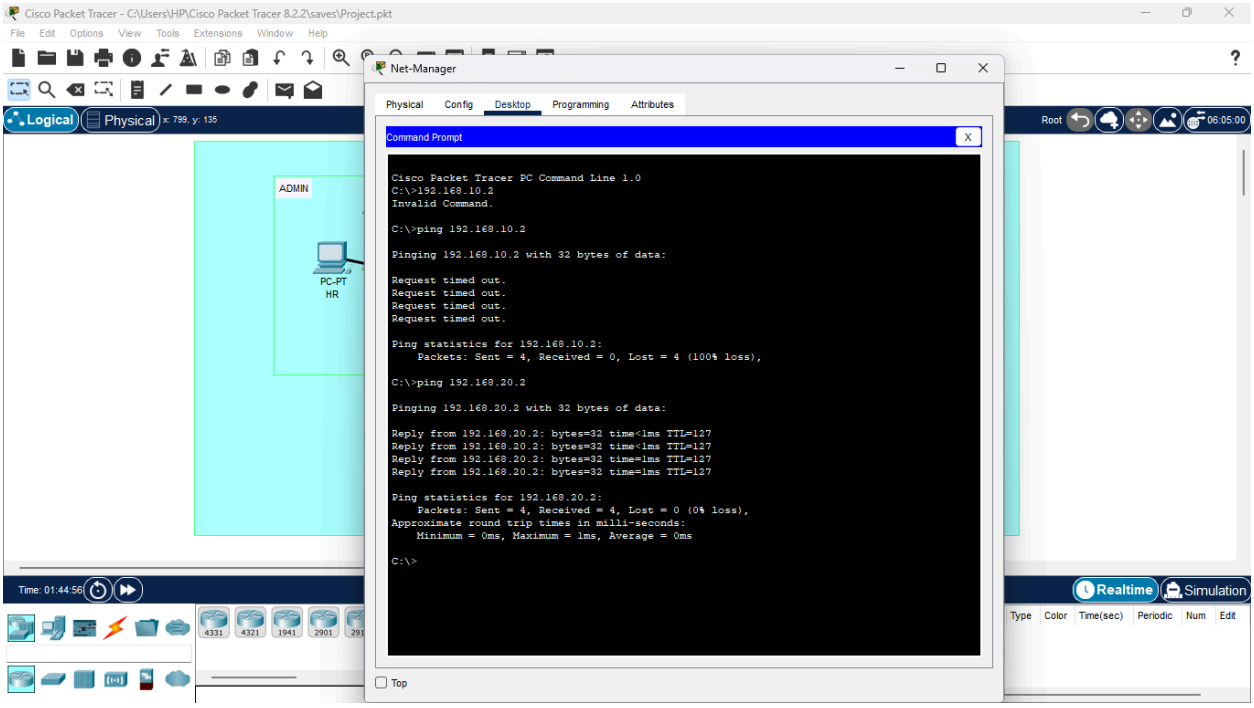
From Sales to Admin Destination unreachable because Sales' outbound rules denies traffic to Admin

From Sales to IT approves traffic from both ends.



From IT to Admin request timed out because Admin outbound rules denies traffic to IT

From IT to Sales approves traffic on both ends



Security Justification Documentation

Rule 1: IT Department Unrestricted Access

bash

```
permit ip 192.168.30.0 0.0.0.255 any
```

Security Justification:

- **Principle of Least Privilege Exception**: IT staff require broad network access to perform administrative tasks, system maintenance, and user support
- **Operational Necessity**: IT needs to reach all systems for monitoring, patching, and troubleshooting
- **Risk Mitigation**: Trust is placed in IT staff with proper authentication and auditing controls
- **Business Continuity**: Ensures IT can respond to incidents across all departments

Rule 2: Admin to Sales Access (Reporting)

bash

```
permit ip 192.168.10.0 0.0.0.255 192.168.20.0 0.0.0.255
```

Security Justification:

- **Business Requirement**: Admin department needs Sales data for financial reporting and analysis
- **Data Flow Control**: Unidirectional access from Admin to Sales only
- **Segmentation**: Prevents Sales from initiating connections back to Admin
- **Compliance**: Supports financial reporting requirements while maintaining department boundaries

Rule 3: Admin to IT Denial

bash

```
deny ip 192.168.10.0 0.0.0.255 192.168.30.0 0.0.0.255
```

Security Justification:

- **Security Boundary**: Prevents administrative staff from accessing IT infrastructure
- **Reduced Attack Surface**: Limits potential lateral movement if Admin workstations are compromised
- **Separation of Duties**: Financial/administrative staff shouldn't have access to IT systems
- **Compliance**: Aligns with security best practices for role-based access

Rule 4: Sales to Admin Denial

bash

```
deny ip 192.168.20.0 0.0.0.255 192.168.10.0 0.0.0.255
```

Security Justification:

- **Data Protection**: Sales staff should not access financial/HR systems in Admin department
- **Confidentiality**: Protects sensitive employee and financial data from sales team
- **Regulatory Compliance**: Prevents unauthorized access to confidential information
- **Trust Boundary**: Maintains separation between customer-facing and internal departments

Rule 5: Sales to IT Access (Support)

bash

```
permit ip 192.168.20.0 0.0.0.255 192.168.30.0 0.0.0.255
```

Security Justification:

- **Operational Support**: Sales team needs IT assistance for system issues
- **Controlled Access**: Limited to IT support systems only
- **Business Enablement**: Ensures sales tools and CRM systems remain operational
- **Monitoring**: IT can provide better support when sales systems can reach IT helpdesk

Rule 6: Default Deny All

bash

```
deny ip any any
```

Security Justification:

- **Implicit Deny**: Explicitly states the default security stance
- **Defense in Depth**: Provides clear logging and monitoring of denied attempts
- **Security Clarity**: Makes the security policy explicit in configuration
- **Audit Trail**: Creates records for security monitoring and compliance

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4. **Secure Remote Access**

Generate RSA crypto key
Define a hostname for the router
Configure a domain name
Enable SSH
Configure SSH usernames and password
Configure AAA model

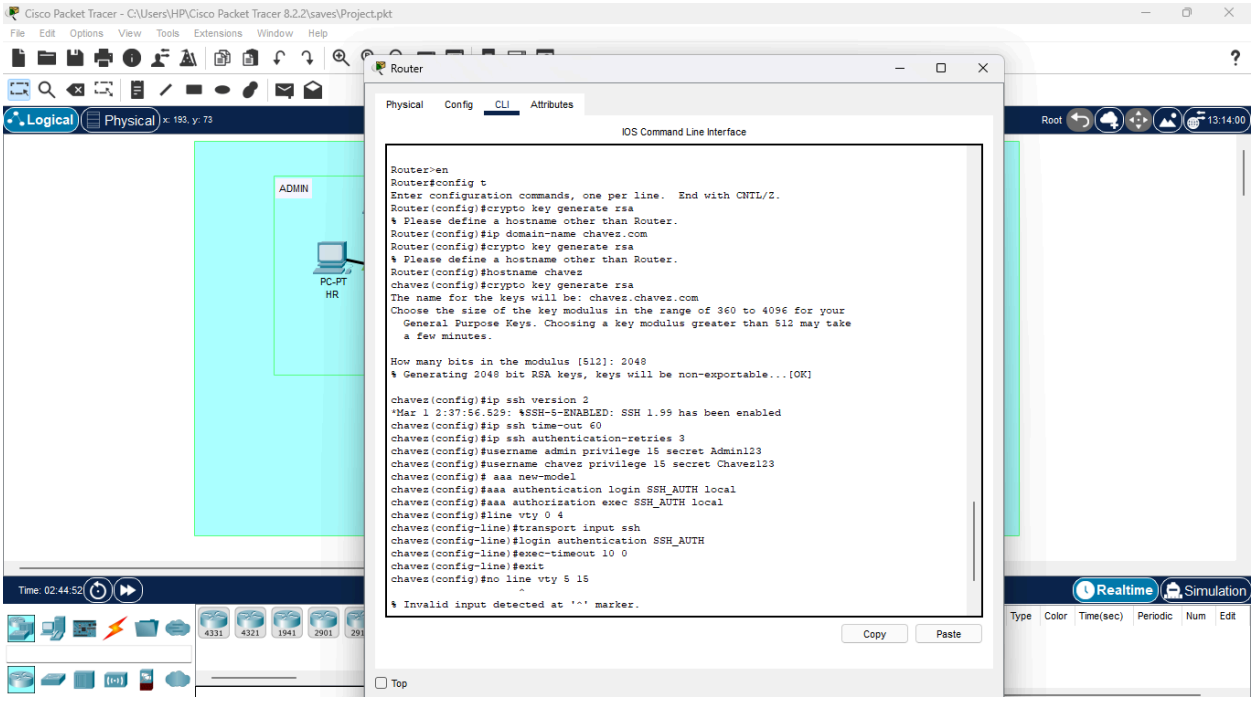
Configure VTY lines for SSH transport And ensure other lines are down to ensure insecure protocols like Telnet are down.

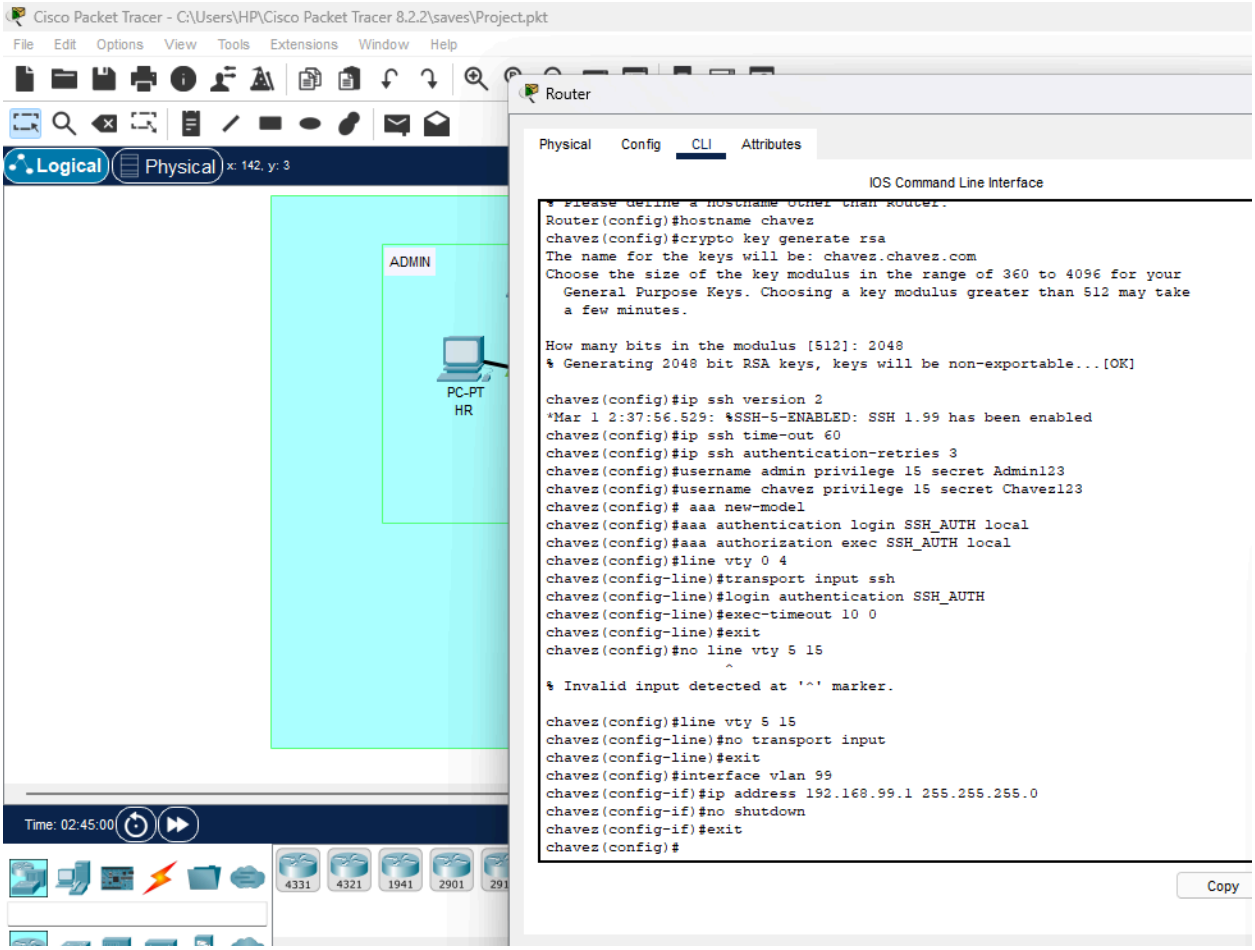
```
ip domain-name chavez.com
hostname chavez
! Generate RSA keys
crypto key generate rsa modulus 1024

! SSH configuration
ip ssh version 2
ip ssh time-out 60
ip ssh authentication-retries 3

! User accounts
username admin privilege 15 secret Admin123
username chavez privilege 1 secret Chavez123


! AAA configuration
aaa new-model
aaa authentication login SSH_AUTH local
aaa authorization exec SSH_AUTH local

! VTY configuration
line vty 0 4
transport input ssh
login authentication SSH_AUTH
exec-timeout 10 0
exit

line vty 5 15
no transport input
exit


interface vlan 99
ip address 192.168.99.1 255.255.255.0
```

no shutdown
exit

Security Benefits Achieved

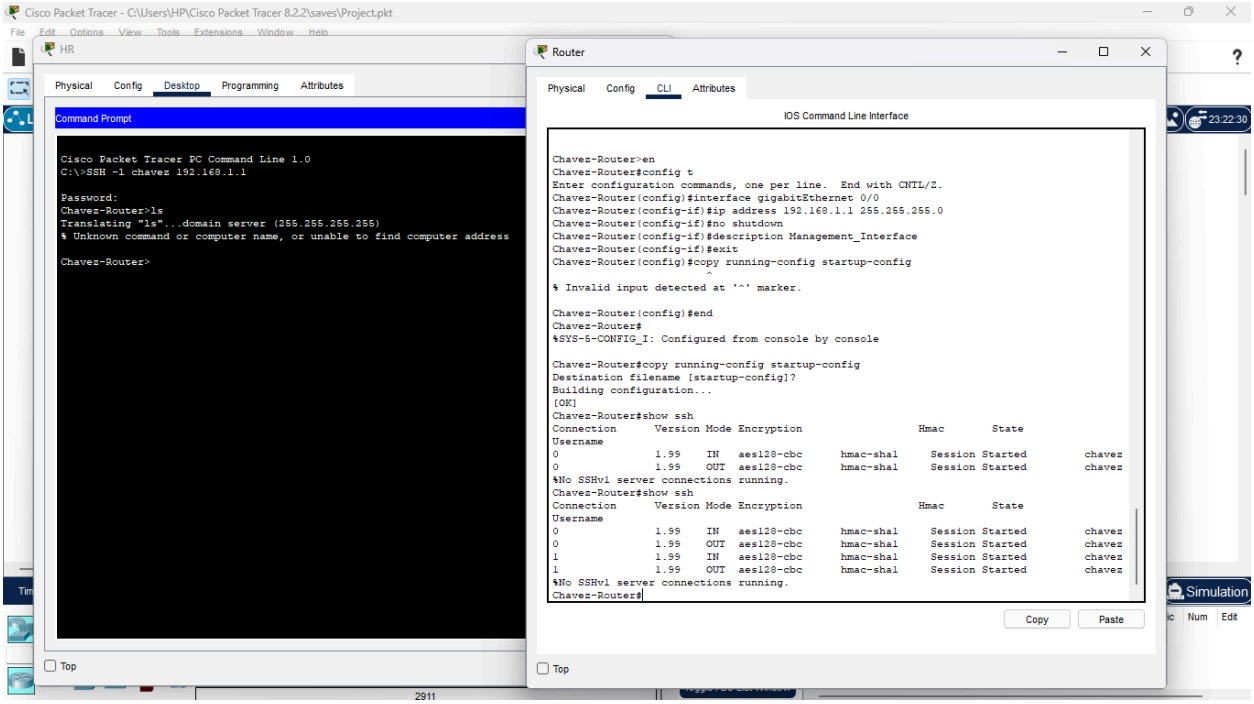
1. **Encrypted Management**: SSH provides encrypted remote access
2. **Telnet Elimination**: Removes clear-text protocol vulnerability
3. **Strong Authentication**: Local user database with encrypted secrets
4. **Access Control**: Restricts management to authorized users only
5. **Session Security**: Timeouts and retry limits prevent brute force





show ssh

On the PC, use the command “SSH -l chavez 192.168.99.1” to connect to Router via SSH


Cisco Packet Tracer - C:\Users\HP\Cisco Packet Tracer 8.2.2\saves\Project.pkt
File Edit Options View Tools Extensions Window Help
HR
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>SSH -l chavez 192.168.1.1
Password:
Chavez-Router>ls
Translating "ls"...domain server (255.255.255.255)
% Unknown command or computer name, or unable to find computer address
Chavez-Router>
Top
Router
Physical Config CLI Attributes
IOS Command Line Interface
Chavez-Router>en
Chavez-Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Chavez-Router(config)#interface gigabitEthernet 0/0
Chavez-Router(config-if)#ip address 192.168.1.1 255.255.255.0
Chavez-Router(config-if)#no shutdown
Chavez-Router(config-if)#description Management_Interface
Chavez-Router(config-if)#exit
Chavez-Router(config)#copy running-config startup-config
^
% Invalid input detected at '^' marker.
Chavez-Router(config)#end
Chavez-Router#
%SYS-5-CONFIG_I: Configured from console by console
Chavez-Router#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
Chavez-Router#show ssh
Connection Version Mode Encryption Hmac State
Username
0 1.99 IN aes128-cbc hmac-sha1 Session Started chavez
0 1.99 OUT aes128-cbc hmac-sha1 Session Started chavez
%No SSHv1 server connections running.
Chavez-Router#show ssh
Connection Version Mode Encryption Hmac State
Username
0 1.99 IN aes128-cbc hmac-sha1 Session Started chavez
0 1.99 OUT aes128-cbc hmac-sha1 Session Started chavez
1 1.99 IN aes128-cbc hmac-sha1 Session Started chavez
1 1.99 OUT aes128-cbc hmac-sha1 Session Started chavez
%No SSHv1 server connections running.
Chavez-Router#
Copy
Paste
Top
23:22:30
Simulation
Num
Edit
2911